

## **Experiment 1**

### **MEASUREMENT WITH WAVEGUIDE BENCH**

# Experiment 1

## Objective:

- Familiarization with the waveguide bench.
- Understanding modulation, demodulation, and detection of microwave signals.
- Measuring standing wave distribution on a slotted line with short-circuit and open-circuit terminations.
- Measurement of guide wavelength  $\lambda_g$  and the frequency  $f$ .
- Plot of dispersion diagram ( $\omega$ - $\beta$  plot).
- Finding the internal broadside dimension,  $a$  of the waveguide.
- Measurement of VSWR of a load.
- Measurement of an unknown impedance.

## Theory:

### Cavity wavemeter:

A calibrated cavity frequency meter (wavemeter) is used in the microwave bench for frequency measurement. The cavity frequency meter consists of a high-Q cylindrical cavity resonator whose effective length may be varied by a movable short-circuit plunger. Resonant frequency of the cavity is determined by its dimensions and the permittivity of the dielectric inside the cavity. The position of the plunger is indicated on a dial. Scale of the dial is calibrated in terms of frequency. The cavity is so designed that it resonates in a particular mode only in the desired frequency range. The cavity is coupled to the waveguide by an iris. The resonant cavity behaves as a narrowband bandstop filter. Power propagation through the waveguide is interrupted when the frequency of electromagnetic wave propagating down the guide matches with the resonant frequency of the cavity and a dip is obtained in the VSWR meter. Be sure to detune the wavemeter after frequency measurement to avoid amplitude fluctuations that may occur when the wavemeter is set to the operating frequency.

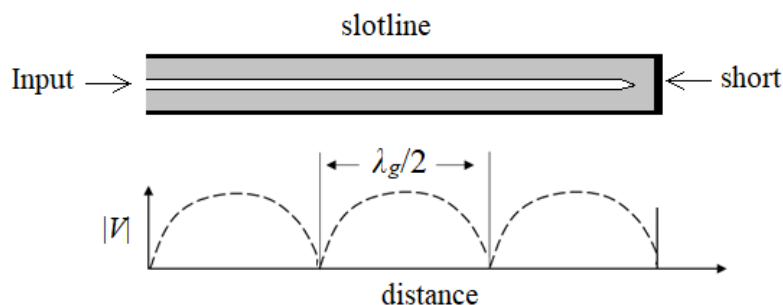


Fig.1 Standing wave distribution inside a waveguide with short-circuit termination.

### Standing wave distribution:

If a transmission line is terminated with an impedance other than its characteristic impedance, the termination is said to be 'mismatched' to the line. Electromagnetic waves traveling down the line are partially or wholly reflected from the termination. Total reflection occurs when the termination impedance is non-dissipative, such as a short circuit, an open circuit, or a purely reactive termination. Standing waves result from two signals of equal wavelength propagating in opposite directions (see Fig. 1). Note that propagating waves and standing waves usually coexist.

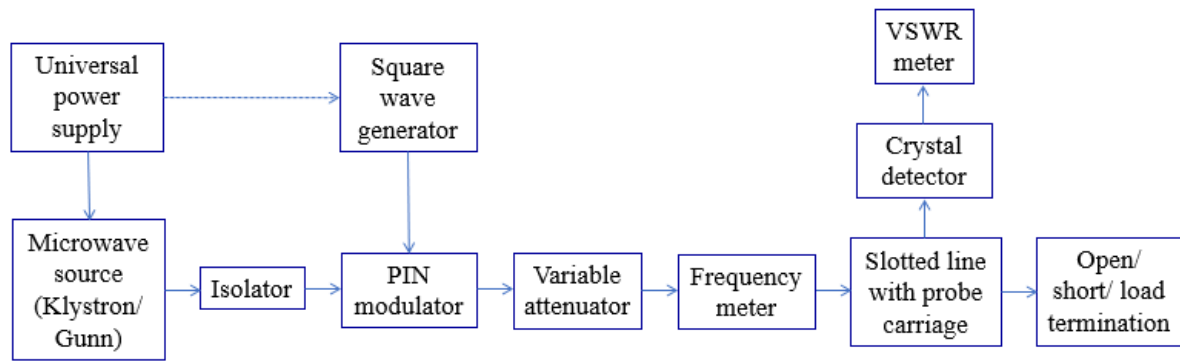
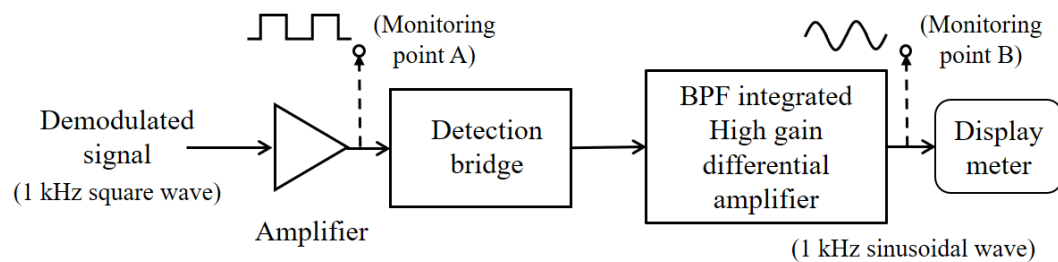


Fig. 2 Test bench set up to measure standing wave distribution.

### Slotted line with probe carriage:

A block diagram of the experimental set up is shown in Fig. 2. The RF source is a square wave modulated Gunn oscillator. The slotted line probe carriage carries a coaxial electric probe to sample the electric field inside the slotted line. The voltage induced in this probe is approximately proportional to the electric field inside the waveguide. A general rule in slotted line measurements is to use minimum penetration of the sampling probe. Otherwise, the power picked up by the probe causes a distortion in the standing wave pattern. The effect on the pattern is greater as the probe penetration is increased. This can be explained by considering the probe as an admittance shunting the line. Therefore, the effect is maximum at the position of  $V_{max}$ . The probe induced voltage is applied to a crystal detector (pin-contact Schottky barrier junction) having square law response. The crystal sits inside the coaxial tube connected to the tunable probe. The crystal output corresponding to the modulation frequency (usually 1 kHz) is fed to a VSWR meter.



(a)



(b)

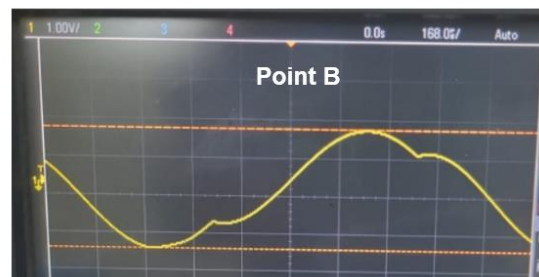


Fig. 3 (a) A simplified block diagram of the VSWR meter and (b) the expected output waveforms at the two monitoring points.

### VSWR meter:

A VSWR meter is a narrowband high gain linear amplifier, usually tuned at a frequency of 1 kHz. A simplified block diagram of a VSWR meter is shown in Fig. 3. Its typical sensitivity is 0.2  $\mu\text{V}$  at 200  $\Omega$  (in some cases 200 k $\Omega$ ) input impedance for full-scale deflection. The reading is proportional to incident power. Since the crystal has a square law characteristic only for small powers, the amplifier should be set at almost full gain so that as small a power as possible gives full scale reading on the meter. This can easily be checked by decreasing the power level with the attenuator and verifying that the power reading (in dB) indicated on the SWR meter drops by the same amount. Alternatively, the power level can be reduced at the source. But, it is usually best to work with a minimum probe depth, and maximum source power to maintain a good signal to noise ratio.

The VSWR meter can be used to measure a standing wave ratio as a direct reading or as a decibel reading to be converted to SWR. VSWR measurement steps are given below.

- Peak the VSWR meter (i.e. move needle to right) by adjusting the modulation frequency of the signal source, if adjustable.
- Peak the reading by tuning the probe detector (impedance tuning by length of the coaxial tube).
- Peak the VSWR meter by moving the probe along the slotted line.
- Adjust GAIN and VERNIER of VSWR meter and/or output power from signal source to obtain a SWR reading of exactly 1.0.
- Move the probe along the line to a voltage minimum (needle moves to left). Do not retune probe or detector once set to 1.0.
- Choose an appropriate scale. For example, if the needle is to the left of 3.2 (out-of-scale) on the 1 to 4 SWR scale, change the RANGE switch to the next 10 dB position above the initial setting. The SWR is then indicated directly on the 3.2 to 10 dB scale. For a reading between 1 and 1.3 on the 1 to 4 SWR scale, change the EXPAND switch to 0.0. This SWR segment is then expanded to full scale and the reading is taken from the 1 to 1.3 dB scale.
- Observe the demodulated ON/OFF signals before and after the 1 kHz narrowband bandpass filter (BPF) on a CRO. These signals can be accessed from behind the VSWR meter. Trace and report the signals.

### Frequency and wavelength:

The distance between successive minima of a standing wave pattern determines the half wavelength  $\lambda_g/2$  on the line. For the TE<sub>10</sub> mode, the guide wavelength  $\lambda_g$  is related to the free space wavelength  $\lambda_0$  by

$$\lambda_g = \frac{\lambda_0}{\left[1 - \left(\frac{\lambda_0}{2a}\right)^2\right]^{1/2}}, \quad (1)$$

which can be expressed as

$$\frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{4a^2}, \quad (2)$$

and the frequency  $f$  is

$$f = \frac{c(\lambda_g^2 + 4a^2)^{1/2}}{2a\lambda_g}, \quad (3)$$

where  $a$  is the larger internal dimension of the guide and  $c$  is the velocity of light in free space. For a lossless line, the phase constant  $\beta$  is

$$\beta = \frac{2\pi}{\lambda_g}. \quad (4)$$

### High VSWR by double minimum method:

The voltage standing wave ratio of a transmission line terminated in a load is defined as

$$VSWR = \frac{V_{\max}}{V_{\min}}, \quad (5)$$

where  $V_{\max}$  and  $V_{\min}$  are the voltage magnitude at the maxima and minima of voltage standing wave distribution. When the VSWR is high ( $\geq 5$ ), the standing wave pattern will have a high maxima and low minima. Since the square law characteristics of a crystal detector is limited to low microwave power, an error is introduced if  $V_{\max}$  is measured directly. This difficulty can be avoided by using the ‘double minimum method’ in which measurements are taken on the standing wave pattern near the voltage minimum. The procedure consists of first finding the value of voltage minima. Next two positions about the position of  $V_{\min}$  are found at which the output voltage is twice the  $V_{\min}$  value. If the detector response follows square law, VSWR is given by

$$VSWR = \left[ 1 + \frac{1}{\sin^2\left(\frac{\pi d}{\lambda_g}\right)} \right]^{1/2}, \quad (6)$$

where  $\lambda_g$  is the guide wavelength and  $d$  is the distance between the two points where the voltage is  $2V_{\min}$ .

### Measurement of unknown impedance:

Assume the rectangular waveguide as a transmission line. Please note that it is possible considering the concept of terminal behaviors (measurable at the ports) and then comparing with a TEM line. Then, the problem is similar to estimate the terminating impedance on a slotted waveguide to finding an unknown load impedance on a transmission line. From the transmission line theory, the input impedance is

$$Z(l) = Z_0 \left[ \frac{Z_L \cos \beta l + j Z_0 \sin \beta l}{Z_0 \cos \beta l + j Z_L \sin \beta l} \right]$$

where  $l$  is the length from the load end,  $Z_0$  is the characteristic impedance of the line,  $\beta = 2\pi/\lambda_g$  is the phase constant, and  $Z_L$  is the load impedance. In terms of normalized impedance (w.r.t.  $Z_0$ ), the variation of impedance is

$$z(l) = \left[ \frac{z_L \cos \beta l + j \sin \beta l}{\cos \beta l + j z_L \sin \beta l} \right], \quad (7)$$

where  $z(l) = Z(l)/Z_0$ , and  $z_L = Z_L/Z_0$ . However, only the voltage standing wave ratio (VSWR) and the locations of maxima and minima can be measured. Therefore, the normalized impedance of an unknown load should be found from this information alone. For this purpose, Smith Chart can be used to calculate the unknown impedance from (a) VSWR meter reading, and (b) the relative distance of voltage maximum and/or minimum.

### Smith Chart:

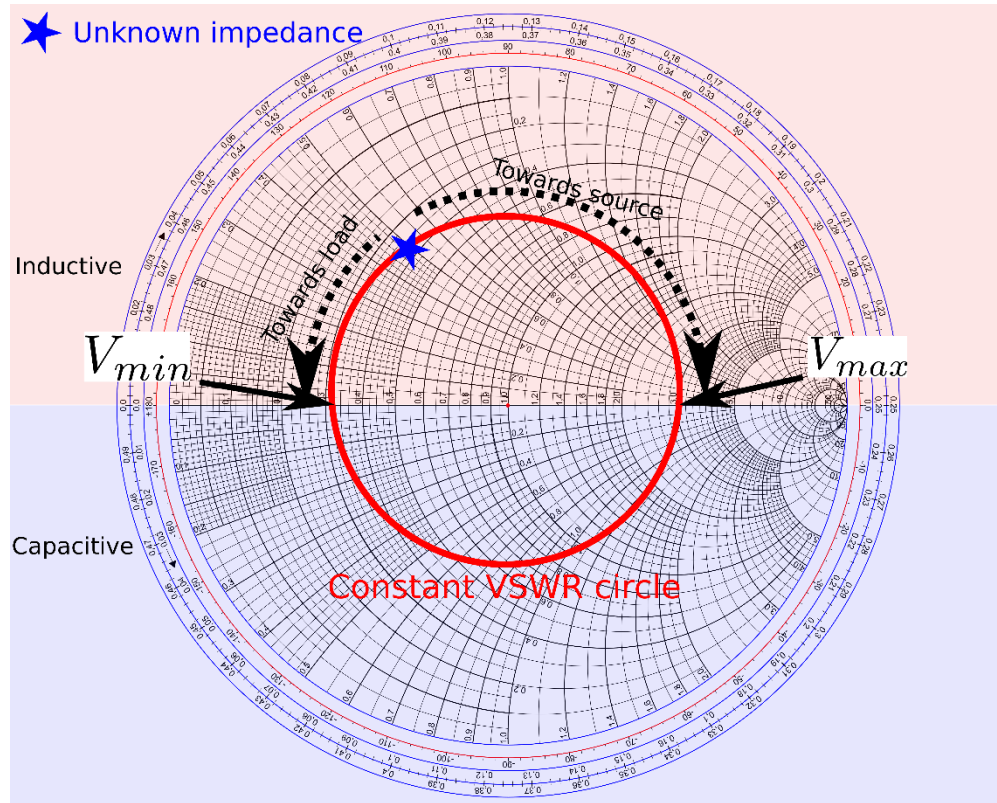


Fig. 4. The unknown impedance point is marked by the star. The waveguide short termination always supports a voltage minimum. The minimum is repeated at an interval of  $\lambda_g/2$ . For a lossless line, the radius of VSWR circle remains constant. Relative distance from the minimum voltage point will give the position of the unknown impedance on the constant VSWR circle. If the impedance is in the upper half, the impedance is inductive, else it is capacitive.

Smith chart maps a complex impedance  $Z(l)$  on a complex reflection coefficient  $\Gamma(l)$  plane. A sample Smith Chart is shown in Fig. 4. Any circle centered at the origin is a constant VSWR circle, as marked in the figure. Since the reflection coefficient varies as  $\Gamma(l) = \Gamma_L e^{-2j\beta l}$ , therefore, as one moves along the length of the line, an impedance point moves on the constant VSWR circle. The angle of this rotation on the constant VSWR circle is given by  $-2\beta l$ . If movement is towards the source, the point moves in clockwise direction. The right intersection of the constant VSWR circle with the x-axis corresponds to the maximum voltage along the line, and the left intersection corresponds to the minimum voltage. Reading from the VSWR meter will determine the radius of the circle, centered at origin, on which the unknown impedance lies. Radius of the circle is  $|\Gamma_L|$  (with largest VSWR circle on the chart normalized to  $|\Gamma_L| = 1$ ), and VSWR is related to the magnitude of reflection coefficient as

$$VSWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}. \quad (8)$$

Relative distance to the voltage minima gives the position of the load impedance on the constant VSWR circle.

## Apparatus:

1. Klystron/Gunn power supply.
2. Klystron Mount (2K25 Reflex klystron) or Gunn mount.
3. Isolator.
4. Cavity wavemeter.
5. Variable attenuator.
6. Waveguide Slotted Section with carriage and untuned probe.
7. Waveguide-to-coaxial adaptors.
8. One detector mount.
9. One shorting plate.
10. Unknown loads and matched termination.

## Procedure (part I):

1. Set up the test bench along with short-circuit termination and other blocks as shown in Fig. 2.
2. Before switching on any instrument familiarize yourself with the operation of each unit and consult instruction manuals, wherever necessary.
3. Set the controls of the power supply for rated electrode voltage of the Klystron/ Gunn.
4. Switch on all the instruments and wait for about 20 minutes before recording any data.
5. Adjust the square wave modulation amplitude so that the detector output is maximum.
6. Tune the probe for maximum sensitivity by adjusting the tuning plunger.
7. Tune the wavemeter until a maximum fall in the square wave output amplitude is obtained. Since the wavemeter is a high Q circuit, you should rotate the wavemeter slowly, otherwise, might miss the minimum power position. Note down the frequency reading. Detune the wavemeter after reading.
8. Switch the probe output of the slotted line to the pre-amplifier, slide the probe carriage along the slotted line and accurately locate the positions of minima on the scale of the slotted line. The minima are best located by finding the positions corresponding to equal but small power levels on either side of the desired minima. The average of the two positions represents the positions of the true minimum. Since, the distance between adjacent minima is equivalent to half the guide wavelength, the value of  $\lambda_g$  is directly obtained.
9. Find the value of  $\lambda_g$  from slotted line and  $f$  from the cavity wavemeter for each setting of the mechanical tuner of the Klystron/ Gunn (8-10 readings over a wide frequency range).
10. In a table, compare calculated  $\lambda_g$  from (1), measured  $\lambda_g$ , calculated  $f$  by (3), measured  $f$  by the wavemeter.
11. Plot  $\beta/k_0$  vs.  $f$ , where  $k_0 = 2\pi/\lambda_0$ .
12. Plot  $1/\lambda_g^2$  vs.  $1/\lambda^2 (= f^2/c^2)$  and hence find the value of  $a$ . Compare the value of thus obtained with the actual value of  $0.900'' \pm 0.002''$  for the WR-90 waveguide.

## Measurement of high VSWR:

1. Connect the load to the end of the slotted line.
2. Locate a position of  $V_{min}$  on the slotted line. Adjust the VSWR meter gain to some reference value, say, 3 dB.

3. Move the probe along the slotted line on either side of  $V_{max}$  so that the reading is 3 dB below the reference i.e. 0 dB. Record the probe positions and obtain the distance between the two positions. Determine the VSWR using equation (6).

### Procedure (unknown impedance, part II)

1. Connect the short termination and tune the microwave bench.
2. Measure frequency and guided wavelength  $\lambda_g$  in the waveguide. This will give you  $\beta = 2\pi/\lambda_g$ .
3. Find the position of the extreme voltage minimum (nearest to the short-circuited waveguide) for the reference. Note down the scale reading. Since the impedance varies periodically along the line, this position can be taken as the virtual position of the load impedance. Also, note down all other minima positions while moving the probe carriage towards the source.
4. Now remove the short and connect the diaphragm along with a matched termination. This forms the unknown impedance at the termination. Note that the load will radiate without the match termination.
5. Because of the load the location of the minimum and maximum voltage now shifts. Starting from right most position, move the probe towards the source. Note down the position of the new voltage minima. Measure the shift in minima (towards the source) as  $\Delta x$ . The shift  $\Delta x$  provides the angle ( $2\beta\Delta x$ ) of rotation on the Smith chart.
6. Repeat the same for other minima positions to take the average value for improved accuracy of measurement.
7. To measure the VSWR, move the probe to the maximum voltage point. Change the variable attenuation to bring the pointer on VSWR meter to 1.
8. Now come to the minimum voltage point. Reading on the VSWR meter at this point is the required VSWR.
9. Plot the constant VSWR circle on the Smith chart.
10. Move on the constant VSWR circle from  $V_{min}$  point on the  $-X$  axis by an angle  $2\beta\Delta x$  (using  $\Delta x$  measured in step-5) in anti-clockwise direction (towards load), to get the unknown impedance. Note that the  $V_{min}$  positions obtained in the first step with the short represent the virtual load positions.
11. Read the normalized resistance and reactance value of this point from the Smith Chart. It provides the normalized impedance of the unknown load.
12. Repeat the above steps for different frequencies and load terminations. Note down your observations.

**Sample questions:**

- If the excitation of the waveguide is changed to TE<sub>20</sub> mode, can we continue with this set up?
- Why is the guide wavelength always greater than the free-space wavelength?
- What is the cut-off wavelength for the TE<sub>10</sub> mode of WR-90 guide?
- Show that only TE<sub>10</sub> mode would propagate in WR-90 guide in the frequency range 8.20 GHz to 12.4 GHz.
- How would you assure yourself that RF power leakage is negligible?
- What is the effect of the probe depth?
- Why is the 10 dB attenuator inserted in the generator side of slotted line?
- Why the slot is cut in the centre of the waveguide not off-centre?
- Why detector is required to have a square law response?
- Why the Klystron/Gunn source is square wave modulated?

**REFERENCES:**

1. E.L. Ginzton, *Microwave Measurement*.
2. D. M. Pozar and E. J. Knapp, *Microwave Engineering Laboratory Notebook*, 2004.